SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1741**

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS:2 **Total Marks:20**

Descriptive Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behaviour and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.

   Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

Code of Conduct:

I S .Arshad ali(192412204) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
S. Arshad ali

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	5
Explanation	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	5
Application	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	4
Organization	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	3
Accuracy	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	2
Clarity	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand	1

ANSWERS

1Q) A smart home automation system is being designed to manage lighting, temperature, and security based on user behaviour and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

ANS:

1. Introduction

An intelligent agent is a system that perceives its environment through sensors and takes appropriate actions using actuators to achieve desired objectives.

In a smart home automation system, sensors detect motion, temperature, light intensity, and time of day. The system uses this information to control lighting, air conditioning, and security devices automatically.

There are four main types of intelligent agents: simple reflex agents, model-based agents, goal-based agents, and utility-based agents.

2. Types of Intelligent Agents

1. Simple Reflex Agent

A simple reflex agent selects actions based only on the current input using predefined condition–action rules.

- Working: It follows an if–then rule.
- Example: If motion is detected in a room, turn on the lights. If no motion is detected for a specified period, turn them off.
- Advantages: Simple, fast, and easy to implement.
- Limitations: It does not remember past states or learn user preferences.

Application: Useful for basic lighting control and immediate security responses.

2. Model-Based Agent

A model-based agent maintains an internal state or model of the environment. It uses current sensor inputs and past information to understand the situation.

- Working: It updates its internal state based on sensor data and previous observations.
- Example: If a person enters a room at night, the system considers the time, previous occupancy, and room conditions before adjusting the lights.
- Advantages: Works effectively in partially observable environments and considers previous states.
- Limitations: Requires additional memory and a more complex system model.

Application: Useful for tracking room occupancy, monitoring environmental changes, and managing temperature based on past conditions.

3. Goal-Based Agent

A goal-based agent selects actions by considering whether they help achieve a specific goal.

- Working: It evaluates possible actions and chooses those that move the system toward its target.
- Example: If the goal is to maintain a comfortable room temperature of 24°C, the agent operates the air conditioner or heater as needed.
- Advantages: Flexible and capable of planning actions to achieve specific objectives.

- Limitations: Planning can require more computation, and it may not effectively compare conflicting goals without additional evaluation criteria.

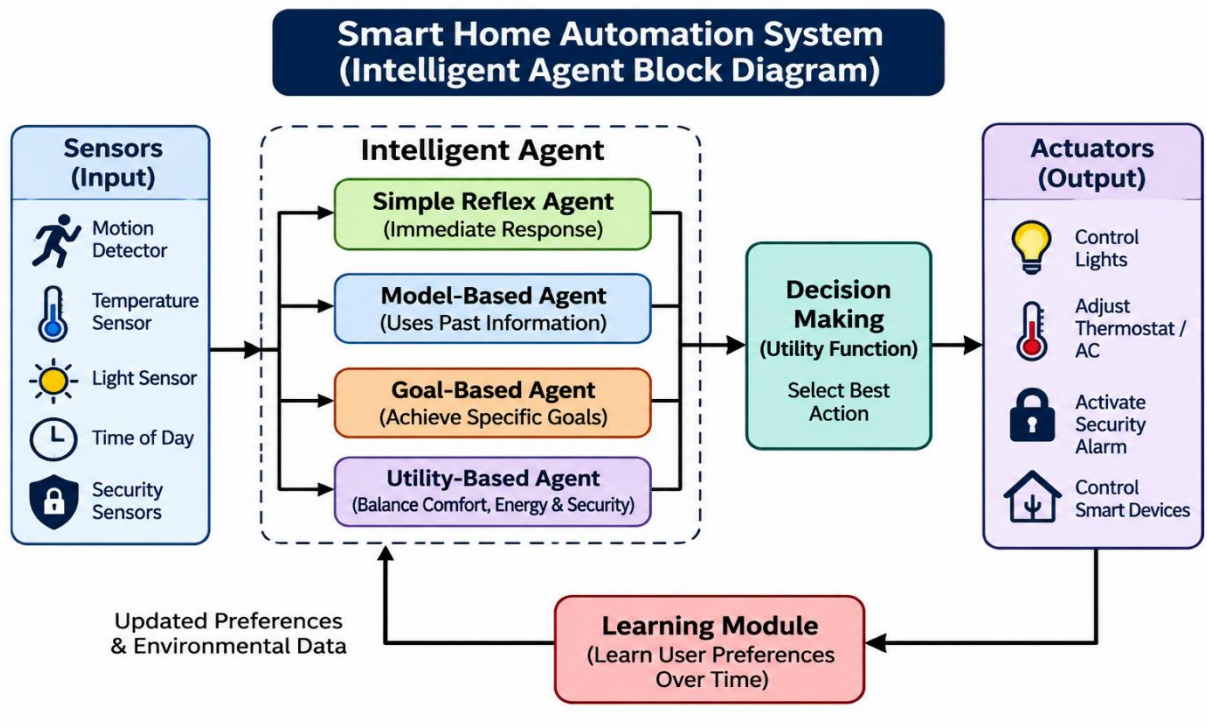
Application: Useful for achieving goals such as maintaining comfort, reducing energy consumption, and ensuring home security.

4. Utility-Based Agent

A utility-based agent evaluates possible actions using a utility function, which assigns a value to each outcome. It chooses the action that provides the greatest overall benefit.

- Working: It compares comfort, energy consumption, safety, and user preferences before selecting an action.
- Example: When the room is empty, the agent may reduce air conditioning to save electricity. When a person returns, it restores the preferred temperature while maintaining security.
- Advantages: Balances multiple objectives and makes effective decisions when goals conflict.
- Limitations: Designing an appropriate utility function is complex and requires reliable data.

Application: Best suited for balancing comfort, energy efficiency, security, and personalized preferences.



5. Conclusion

In conclusion, each intelligent agent has a different role in smart home automation. Simple reflex agents provide quick responses, model-based agents maintain environmental information, goal-based agents achieve specific objectives, and utility-based agents balance competing requirements.

Q2) A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

ANS:

1. Introduction

Uninformed search strategies are search techniques that explore possible paths without using additional knowledge about the distance to the goal. A robot placed in an unknown maze can use these strategies to explore the environment, avoid dead ends, and find a path to the exit.

Two important uninformed search strategies are Uniform Cost Search (UCS) and Iterative Deepening Search (IDS).

2. Uniform Cost Search (UCS)

Uniform Cost Search expands the node with the lowest total path cost from the starting position. It uses a priority queue to select the next position to explore.

Working:

1. Start from the robot's initial position.
2. Add the available neighboring positions to a priority queue.
3. Select the position with the lowest accumulated path cost.
4. Continue exploring until the exit is found.
5. Return the path with the minimum total cost.

Example: If different maze paths have different movement costs, UCS finds the least-cost path to the exit.

Advantages:

- Finds an optimal solution when movement costs are non-negative.
- Complete when each step cost is bounded below by a positive value.
- Suitable for mazes with different movement costs.

Disadvantages:

- Requires large memory to store explored positions and the frontier.
- Can take considerable time when the maze is large.

3. Iterative Deepening Search (IDS)

Iterative Deepening Search repeatedly performs depth-limited search, increasing the depth limit each time until the exit is found.

Working:

1. Start searching with a depth limit of 0.
2. If the exit is not found, increase the limit to 1.
3. Repeat the search with limits 2, 3, 4, and so on.
4. Stop when the exit is found.

Example: The robot first explores positions at depth 0, then depth 1, then depth 2, gradually exploring deeper parts of the maze.

Advantages:

- Uses less memory than UCS.
- Complete for a finite branching factor when a solution exists at finite depth.
- Finds a shallowest solution when all steps have equal cost.

Disadvantages:

- Repeats exploration of previously visited positions.
- Can waste time when the maze is deep.
- Does not guarantee the least-cost path when movement costs differ.

4. Time and Space Complexity

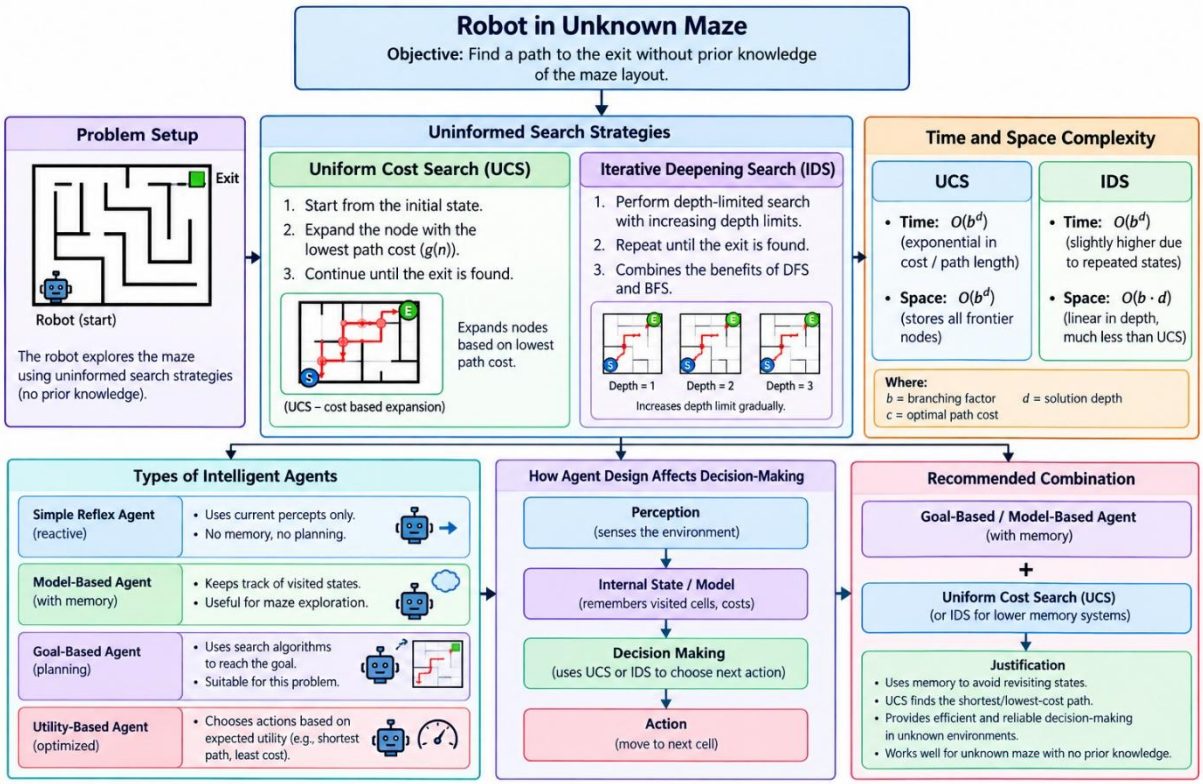
Let \(b\) be the branching factor, \(d\) the depth of the shallowest solution, and \(C^*\) the optimal solution cost.

Feature	**UCS**	**IDS**
Time complexity	Exponential in the worst case; often expressed as \(O(b^{1+\lfloor C^*/\epsilon\rfloor})\) for step costs at least \(\epsilon>0\)	\(O(b^d)\)
Space complexity	Exponential; stores the frontier and explored nodes	\(O(bd)\)
Completeness	Yes, if step costs are bounded below by a positive value	Yes, for finite branching and a finite-depth solution
Optimality	Yes, for non-negative costs under standard assumptions	Yes, when all step costs are equal
Best use	Different movement costs	Equal-cost movements and limited memory

5. Relationship with Types of Intelligent Agents

Different intelligent agents affect how the robot perceives the maze and selects actions.

- Simple Reflex Agent: Uses current sensor input and fixed rules, such as turning left when a wall is detected. It is fast but may repeatedly visit the same locations.
- Model-Based Agent: Maintains an internal map of explored areas and remembers walls, dead ends, and previously visited positions. This helps the robot avoid unnecessary exploration.
- Goal-Based Agent: Selects actions that move the robot toward the goal of reaching the exit. It can use UCS or IDS to find a path.
- Utility-Based Agent: Evaluates paths based on factors such as distance, time, battery usage, and safety, selecting the path with the greatest overall benefit.



6. Conclusion

UCS and IDS are useful uninformed search algorithms for maze-solving robots. UCS is preferable when finding the minimum-cost path is important, whereas IDS is suitable when memory is limited and all movements have equal cost. Therefore, a model-based and goal-based agent combined with UCS is generally the better choice for reliable, cost-efficient maze navigation.